FYP Proposal

AURELIA — Gold Marketplace Platform

BSCS 7th / IET 5th
Proposal (FYP–I)

Cover Information

Project Title	AURELIA — Gold Marketplace Platform
Program / Semester	BSCS 7th / IET 5th
FYP Stage	Proposal (FYP–I)

Team Member

Team Member	Registration No.	Email	Phone

Supervisor

Supervisor	Department	Email	Phone

Co–Supervisor / Industry Mentor (optional)

Name	Organization	Email	Phone

1 Abstract (200–300 words)

AURELIA is a cross–platform marketplace application focused on gold commerce. Buyers and sellers can browse, buy, sell, and auction gold–related listings (bars, coins, jewellery, and investment gold) through a Flutter client backed by a Node.js modular–monolith API.

The platform problem addressed by the implemented system is fragmented, low–trust digital gold trading: users need authenticated accounts, verified sellers or stores, structured gold attributes (purity/karat, weight units, making charges, hallmarks, certificates), searchable listings, and communication channels before completing a trade. AURELIA centralizes these concerns in one product surface branded as a Gold Marketplace.

Technically, the backend is an Express 5 + TypeScript modular monolith using MySQL for persistence, Redis for caching (with in–memory fallback), JWT–based authentication, Zod validation, and Socket.io for realtime chat/presence/auction rooms. Marketplace domains plug in through a registry interface; the active marketplace module in the current codebase is `gold`. Shared platform modules already present include authentication (including OTP, MFA, OAuth, passkeys, and device security), KYC/verification, catalog and listings, auctions, search/filters/maps, favorites and comparison, chat and calls, notifications, subscriptions and payments, reviews, advertisements, AI assistance endpoints, moderation, and risk/fraud–related services.

Gold–specific capabilities in code include store applications, purity/rate quoting, rate history and forecast endpoints, authenticity/assessment helpers, buy/delivery confirmation flows, bidding, certificates/hallmarks, and verification requests. The Flutter client exposes corresponding product flows for auth, listings, search, maps, chat, payments, seller tools, and gold stores.

Expected outcomes of the project are a working end–to–end gold marketplace client and API with the modules listed above. Property and Vehicles marketplace schemas exist historically and are currently deactivated in the database; they are out of active API scope unless later restored.

2 Problem Significance & Context

2.1 Background & Motivation

Gold is traded as bars, coins, jewellery, and investment metal. Digital listing, pricing (rate per gram/tola/ounce, making charges, wastage), hallmarking/certificates, and buyer–seller communication are all required for a usable marketplace. The AURELIA product surfaces these needs under a single app tagline: “Buy, sell and auction gold with confidence.”

Stakeholders reflected in the system design include:

- End users (guests and registered buyers/sellers)
- Gold stores / businesses applying through store workflows
- Platform operators via admin, moderation, support, and analytics modules

2.2 Problem Statement

Buyers and sellers lack a unified, trust–oriented digital channel to list, price, verify, search, communicate about, and auction gold items with marketplace–specific attributes (purity, weight, hallmarks, certificates, and live/quoted rates).

2.3 Scope & Assumptions

In scope (confirmed in current codebase):

- Flutter client branded AURELIA (Gold Marketplace)
- Modular–monolith API with active `gold` marketplace module
- Platform capabilities: auth/security, KYC/verification, listings, auctions, search/filters/maps, favorites/compare, chat/calls, notifications, subscriptions/payments, reviews, ads, AI endpoints, moderation/risk, seller/support/admin
- Gold features: stores, catalog/brands/hallmark authorities, rates and quotes, purity standards, authenticity/assessment helpers, buy/delivery, auctions/bids, certificates, verification requests

Out of scope / inactive today:

- Property and Vehicles marketplace API modules (schema/seed rows exist; deactivated; no active backend marketplace packages)

Assumptions / constraints visible in implementation:

- Default AI provider is heuristic/offline; an OpenAI–compatible driver is optional via configuration
- Payment gateways without configured drivers fall back to manual settlement paths
- Gold authenticity helpers in code are not chemical assays

3 Literature Review & Gap Analysis

No literature sources are recorded in the repository yet — matrix left blank.

Ref#	Year	Domain	Method	Dataset	Findings	Gap	Relevance

4 Objectives and Scope

#	Objectives	Data Source / Tooling / Technique
1	Deliver a Flutter client for browsing and trading gold listings (buy/sell/auction) with auth and localization flows.	Flutter, go_router, signals, dio, socket_io_client, Drift/SQLite
2	Provide a modular–monolith API with a pluggable marketplace registry and an active Gold module.	Node.js $\geq$ 20, Express 5, TypeScript, Zod
3	Support gold commerce data: purity/weight, rates/quotes, stores, hallmarks/certificates, auctions/bids.	MySQL (`marketplace`), gold migrations/services
4	Provide shared platform services: search/filters/maps, chat/calls, notifications, payments/subscriptions, reviews, KYC, moderation/risk, AI helper endpoints.	MySQL, Redis (cache), Socket.io, JWT, provider adapters

5 Methodology & Technical Approach

5.1 System/Algorithm Design

[Insert system architecture diagram here]

High–level architecture present in the repository:

- **Client:** Flutter app (`frontend/`) — routing, feature modules, local persistence, realtime socket client
- **API:** Express modular monolith (`backend/src/`) — platform routers + marketplace registry mount at `/gold`
- **Data:** MySQL schema via numbered migrations; Redis cache; Socket.io rooms for user/conversation/market
- **Providers:** AI (heuristic default / OpenAI–compatible), payment gateway adapters, mailer

Module	Description	Inputs	Outputs
Flutter client	AURELIA UI and local state	User actions	API/socket calls
Auth & security	Register/login/OTP/MFA/OAuth/passkeys/devices	Credentials/tokens	Session JWT
Verification/KYC	Document/verification workflows	User/business docs	Verification status
Listings & auctions	Create/browse/bid on listings	Listing payloads, bids	Listing/auction records
Search/filters/maps	Discovery over catalog/listings	Queries, geo filters	Result sets
Chat/calls/notifications	Real-time communication	Messages/events	Delivered events
Payments/subscriptions	Billing via gateway adapters	Payment intents	Settlement status
AI module	Description, search interpret, price helpers, forecasts, checks	Text/listing context	Heuristic/model responses
Risk/moderation	Trust and abuse controls	Signals/cases	Risk outcomes
Gold marketplace	Stores, rates, purity, authenticity helpers, certificates	Gold attributes	Quotes, gold APIs
Admin/seller/support	Operational dashboards	Operator actions	Config/case updates

5.2 Data Plan (collection, licensing, privacy, preprocessing)

Dataset	Source/License	Features/Labels	Preprocessing
Marketplace DB schema/seeds	Project migrations & seed scripts	Platforms, users, listings, gold catalogs/rates, chat, payments, risk tables	SQL migrations; seed scripts
User–generated listings & media	App users (runtime)	Gold attributes, images/uploads	Zod validation; upload pipeline

5.3 Baselines & Experimental Plan

No evaluation baselines or target metrics are defined in the repository.

Baseline	Description	Hyperparams	Expected Outcome	Metrics

5.4 Tools & Resources

Resource/Tool	Purpose
Flutter / Dart SDK	Cross–platform client
Node.js + TypeScript + Express 5	HTTP API
MySQL (`mysql2`)	Primary datastore
Redis	Caching (in–memory fallback available)
Socket.io	Realtime chat/presence/auctions
JWT + Zod	Auth tokens and request validation
Helmet, CORS, rate–limit, Pino	Security headers, CORS, throttling, logging
Nodemailer	Email delivery
Drift / SQLite (client)	Local persistence on device
flutter_map / geolocator	Maps and location
AI provider adapters	Heuristic default; optional OpenAI–compatible
Payment provider adapters	Stripe/PayPal/bank/regional/manual paths as coded

6 Feasibility, Timeline & Risks

6.1 Work Breakdown Structure (WBS) & Timeline

Calendar dates and ownership are not recorded in the repository.

Task/Work Package	Start	End	Days	Deps	By
Platform core (auth, users, verification, catalog)					
Listings, auctions, search, maps, favorites/compare					
Chat, calls, notifications (Socket.io)					
Payments, subscriptions, ads, reviews					
AI, moderation, risk modules					
Gold marketplace module + Flutter gold/store flows					
Admin / seller / support dashboards					

Paste a Gantt chart below.

6.2 Risk Register

Risk	P (L/M/H)	I (L/M/H)	Mitigation Plan
Payment gateways unconfigured in deployment			Use configured adapters or manual settlement path already present in code
AI quality depends on provider configuration			Keep heuristic driver as fallback; enable OpenAI–compatible driver when keys exist
Property/Vehicles scope confusion vs Gold–only API			Treat deactivated marketplaces as out of scope unless modules are restored
Trust features (AML/authenticity) partial/placeholder in places			Document limits; avoid claiming chemical assay or licensed AML without wiring

7 Ethics, SDGs, and IP Considerations

7.1 Ethics & Data Protection

The system includes authentication, device/security controls, KYC/verification document workflows, and risk/moderation modules that process sensitive user and business data. Deployment must protect credentials, personal data, and uploaded documents according to applicable institutional and legal rules.

Institutional ethics-board process / consent forms: unknown — left blank.

7.2 Societal Impact

SDG mapping not specified in the repository — left blank.

7.3 IP & Publication

Patent/paper plan not specified in the repository — left blank.

Third–party libraries used by the project retain their own licenses (Flutter/Dart packages, Node npm packages).

8 Expected Outcomes & Deliverables

Deliverable	Description	Acceptance Criteria	Due	Owner
Flutter client	AURELIA app with gold marketplace UX flows	Builds/runs; core gold browse/auth/list flows usable		
Backend API	Modular monolith with `/gold` and platform routes	Health OK; key modules respond per routes		
Database migrations/seeds	Marketplace schema and gold data model	Migrations apply cleanly		
Gold commerce features	Stores, rates/quotes, auctions, certificates/verification hooks	Gold endpoints and client screens operate end–to–end		
Trust & ops surfaces	KYC, moderation/risk, admin/seller/support	Modules reachable; basic workflows executable		

9 Evaluation Plan & Success Criteria

Numeric targets are not defined in the repository.

Metric	Dataset/Testbed	Target/Threshold
API health / DB connectivity	Local or staging environment	
Auth flows (register/login/OTP/MFA as enabled)	Test accounts	
Gold listing create/search/view	Seeded + test listings	
Auction bid path	Test auction listing	
Chat message delivery (Socket.io)	Two test clients	

Map metrics to the objectives in Section 4 after targets are agreed with the supervisor.

10 Project Management & Meetings

10.1 Meeting Cadence

Weekly with Supervisor; Meeting Log maintained (date–wise & task–wise).

10.2 Roles & Responsibilities

Area/Task	Member(s)	Supervisor/Co–Sup Role

10.3 Annex E

Team Contribution Sheet will be updated and submitted before each evaluation.

11 Compliance Checklist (Attach at Submission)

Item	Yes/No	Notes
1. Abstract within 200–300 words		Drafted from codebase facts
2. Problem statement precise; significance with evidence		Product-feature based; no external market statistics in repo
3. Gap articulated vs. literature (matrix attached)		Literature matrix blank — sources unknown
4. Objectives		Filled from implemented scope
5. Methodology sound; baselines defined; evaluation plan ready		Methodology filled; baselines/targets blank
6. Data plan		Schema/seeds and UGC noted
7. WBS/timeline realistic; risks with mitigations		Tasks listed; dates blank; risks from known code limits
8. SDG/Ethics		Ethics notes partial; SDG blank
9. Evaluation Plan added		Metric names only; thresholds blank
10. Annex E (Team Contribution Sheet) attached		

12 Declarations & Signatures

We declare that the information provided is accurate and that we will adhere to NSU FYP rules & guidelines.

Team Member	Signature	Date

Supervisor Endorsement: _____ Date: _____/_____/_____

FYP Coordinator Review: _____ Date: _____/_____/_____